

1. Administrative & Study Identification

Full Study Title: A Randomized, Open-Label Phase 2 Clinical Trial of BMS-986012 in Combination with Carboplatin, Etoposide, and Nivolumab as First-Line Therapy in Extensive-Stage Small Cell Lung Cancer **Short Title/Acronym:** CA001-050 **Protocol Number:** CA001-050 **NCT Number:** NCT04702880 **Posting ID:** 947832965169520676 **Sponsor Name / Company:** Bristol Myers Squibb **Investigational Product:** BMS-986012 (atigotatug), Nivolumab (Opdivo), Carboplatin, Etoposide **Phase of Development:** Phase II **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate that treatment with BMS-986012 in combination with carboplatin, etoposide, and nivolumab will have acceptable safety and tolerability, and will improve progression-free survival (PFS) compared with carboplatin, etoposide, and nivolumab alone in newly diagnosed participants with extensive-stage small cell lung cancer (ES-SCLC). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DOR), tumor fucosyl-GM1 and PD-L1 expression, and BMS-986012 immunogenicity.

2.2 Background & Rationale Fucosyl-GM1 (Fuc-GM1) is a monosialoganglioside highly expressed in the majority of SCLC cases but with limited expression in normal tissues. BMS-986012 is a first-in-class, fully human IgG1 monoclonal antibody designed to bind Fuc-GM1 with high affinity, eliciting antibody-dependent cell-mediated cytotoxicity (ADCC). This study aims to evaluate whether adding BMS-986012 to the standard first-line chemoimmunotherapy backbone (nivolumab plus carboplatin and etoposide) will boost ADCC and yield superior progression-free and overall survival benefits for ES-SCLC patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Nivolumab + Carboplatin + Etoposide) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 120 patients **Number of Sites:** Global, Multicenter **Estimated Study Start:** March 2021 **Estimated Primary Completion:** December 2022

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically documented extensive-stage small cell lung cancer (ES-SCLC) and extensive-stage disease (American Joint Committee on Cancer, 8th edition, Stage IV [T any, N any, M1a, M1b, or M1c], or T3-4 due to multiple lung nodules that are too extensive or tumor or nodal volume that is too large to be encompassed in a tolerable radiation plan). 3. At least 1 measurable lesion by computed tomography (CT) or magnetic resonance imaging (MRI) per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1). 4. Eastern Cooperative Oncology Group performance status (ECOG PS) 0 or 1. 5. Mandatory provision of an archived tumor specimen (fresh tumor biopsy from any disease site required for PET tracer sub-study). 6. Adequate hematologic and end organ function. 7. Must agree to follow specific methods of contraception, if applicable.

4.2 Exclusion Criteria 1. Prior chemotherapy, radiation therapy, or biologic therapy for SCLC (previously treated LS-SCLC participants are excluded). 2. Symptomatic brain or other central nervous system (CNS) metastases. 3. History of idiopathic pulmonary fibrosis, drug-induced pneumonitis, idiopathic pneumonitis, organizing pneumonia, or evidence of active pneumonitis on screening chest CT scan. 4. Active, known or suspected autoimmune disease, inflammatory disorder, or paraneoplastic autoimmune syndrome requiring systemic treatment. 5. Grade ≥ 2 peripheral sensory neuropathy at study entry. 6. Women who are pregnant or breastfeeding. Japan only: participation in the study is not allowed even if breastfeeding is suspended. 7. Significant uncontrolled cardiovascular disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: BMS-986012 administered in combination with standard doses of nivolumab, carboplatin, and etoposide. Control arm receives standard-of-care nivolumab + carboplatin + etoposide. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Administered until disease progression, unacceptable toxicity, or withdrawal of consent.

Etoposide Drug Profile: * **Drug Preferred Name:** etoposide * **ATC Code:** L01CB01 * **Trade Names:** Etopophos, Vepesid, Toposar

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-induced toxicities and ES-SCLC

symptoms. **Prohibited:** Concurrent systemic therapies for cancer, other investigational agents, and immunosuppressive medications that may interfere with the immune-oncology agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (Archived/Fresh), Baseline Imaging (CT/MRI), ECOG PS
Baseline	Day 0	Randomization, First Dose of Investigational Product(s), Safety Labs
Interim	Protocol-defined cycles	Safety Labs, Efficacy Assessment (RECIST v1.1 imaging), PK/PD sampling, AE monitoring (focus on pruritus)
End of Study	Follow-up	Final Vital Signs, Survival Follow-up, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Incidence of adverse events (AEs) 2. Incidence of serious adverse events (SAEs) 3. Incidence of AEs leading to discontinuation

7.2 Secondary & Exploratory Endpoints * Progression-free survival rate (PFSR) * PFS by investigator based on RECIST v1.1 criteria * Overall Survival (OS) * Objective Response Rate (ORR) * Duration of Response (DOR) * Tumor fucosyl-GM1 and PD-L1 expression, and BMS-986012 immunogenicity profiling

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events per CTCAE criteria. Special attention given to immunologic AEs and pruritus, a notable AE associated with the anti-Fuc-GM1 class. **Serious Adverse Events (SAEs):** Routine 24-hour reporting via the sponsor's pharmacovigilance pathway. **Data Monitoring Committee (DMC):** An independent monitoring committee oversees the safety and risk-benefit profile throughout the study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Population for efficacy endpoints; Safety Population for safety assessments. **Sample Size Justification:** Trial powered to assess meaningful clinical signals (hazard ratio improvements) in PFS and OS compared to the active control arm. **Handling of Missing Data:** Time-to-event endpoints (PFS, OS) subject to standard right-censoring methods at the date of last contact or cutoff.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular clinical site monitoring visits to ensure protocol adherence, accurate data entry, and patient safety. **Audit Trail:** Validated eCRF capturing systems with full audit trails, strict data clarification query resolution processes, and central imaging review.

11. Study Identifiers & Governance

Primary ID: CA001-050 **Registry Number:** NCT04702880 **Posting ID:** 947832965169520676 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study of BMS-986012 in Combination With Carboplatin, Etoposide, and Nivolumab as First-line Therapy in Extensive-stage Small Cell Lung Cancer **Official Regulatory Title:** A Randomized, Open-Label Phase 2 Clinical Trial of BMS-986012 in Combination with Carboplatin, Etoposide, and Nivolumab as First-Line Therapy in Extensive-Stage Small Cell Lung Cancer **Phase ID:** PHASE2 **Criteria Type:** PHASE_REQUIREMENT **Source Level:** 5

12. Clinical Framework (Extensive-Stage SCLC Specific)

Disease Area: Extensive-stage Small Cell Lung Cancer **Name:** Extensive Stage Small Cell Lung Cancer **Preferred Name:** Small cell lung cancer extensive stage **Preferred UMLS Name:** Extensive Stage Small Cell Lung Carcinoma **Semantic Type:** Neoplastic Process **Definition:** Small cell lung carcinoma which has spread beyond one hemi-thorax and the regional lymph nodes. **MeSH Code:** D009369 **SNOMED ID:** 708064003 **SNOMED Hierarchy:** 188554009|Extensive stage small cell lung carcinoma|Extensive stage small cell lung carcinoma (disorder) **Medical Methodology:** Chemoimmunotherapy combined with targeted Monoclonal Antibody therapy **Optimization Target:** Improvement of ADCC, overall survival, and progression-free survival

13. Study Procedures & Methodology

Management Pathway: First-line systemic anti-cancer therapy for newly diagnosed ES-SCLC patients **Education Protocol (HCP):** Investigator meetings and site initiation visits detailing the administration of the four-drug regimen and specific immune-related AE management. **Education Protocol (Patient):** Informed consent process regarding the addition of a novel monoclonal antibody to a standard-of-care regimen. **Quality Audit Mechanism:** Independent central review of radiographic imaging and Sponsor-led clinical quality assurance audits.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or death (PFS/OS tracked over subsequent years). **Onsite Visit Cadence:** Aligned with infusion cycles (e.g., Q3W/Q4W as per standard nivolumab + chemo regimens). **Remote Monitoring Frequency:** Central safety and data monitoring conducted on a rolling basis. **Checklist Review Interval:** Regular tumor assessments (CT/MRI) per protocol schedule (typically every 6 to 8 weeks in ES-SCLC).

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				